Chapter 3: Fundamentals of Crystallography

ISSUES TO ADDRESS...

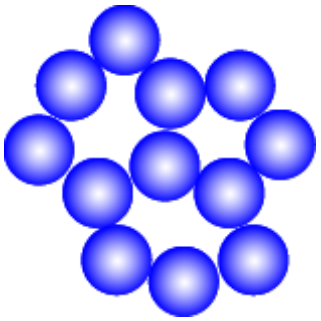
- What is the difference in atomic arrangement between crystalline and noncrystalline solids?
- How are crystallographic directions and planes named?

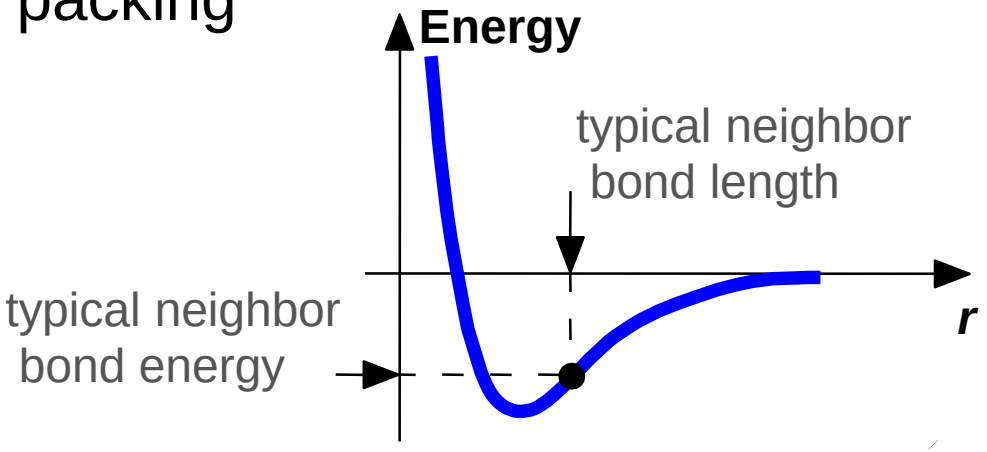
Introduction

- Within this framework, concepts of crystallinity and non-crystallinity are introduced
- The three common crystal structures found in metals are then detailed, along with the scheme by which crystallographic points, directions and planes are expressed
- Single crystals, polycrystalline materials, and non-crystalline materials are considered
- Another section briefly describes how crystal structures are determined experimentally using x-ray diffraction techniques

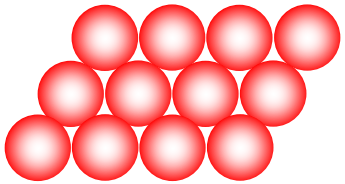
Energy and Packing

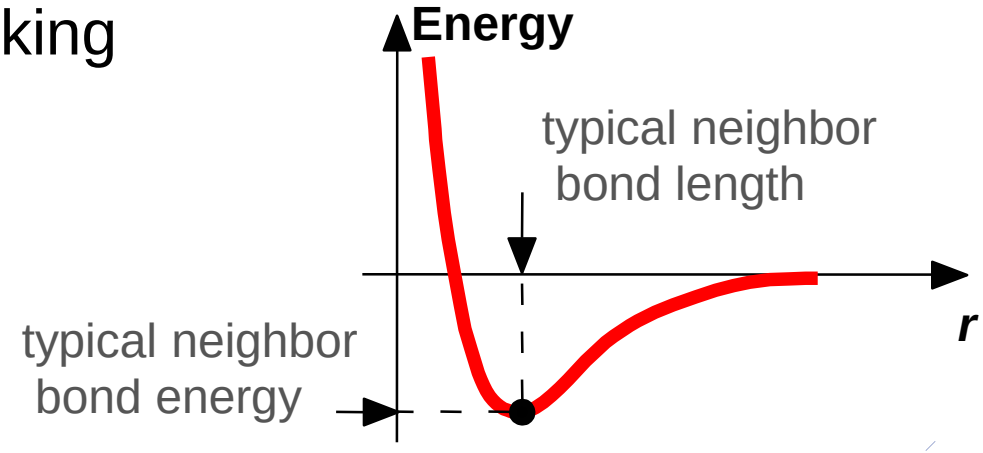
- Non dense, random packing





- Dense, ordered packing



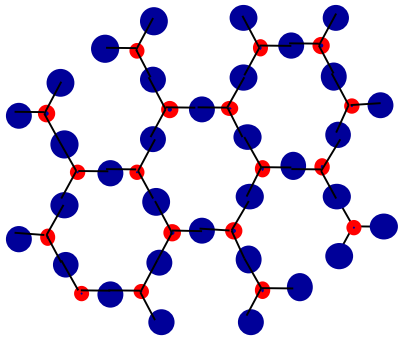


Dense, ordered packed structures tend to have lower energies.

Materials and Packing

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of: -metals
  -many ceramics
  -some polymers



crystalline SiO_2

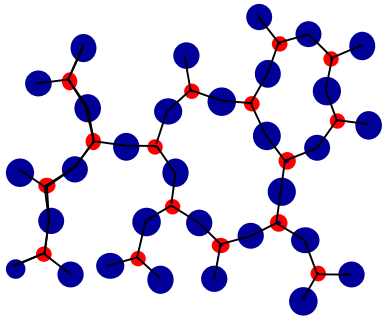
Adapted from Fig. 3.11(a), *Callister & Rethwisch 9e.*

• **Si** • **Oxygen**

Noncrystalline materials...

- atoms have no periodic packing
- occurs for: -complex structures
  -rapid cooling

"Amorphous" = Noncrystalline



noncrystalline SiO_2

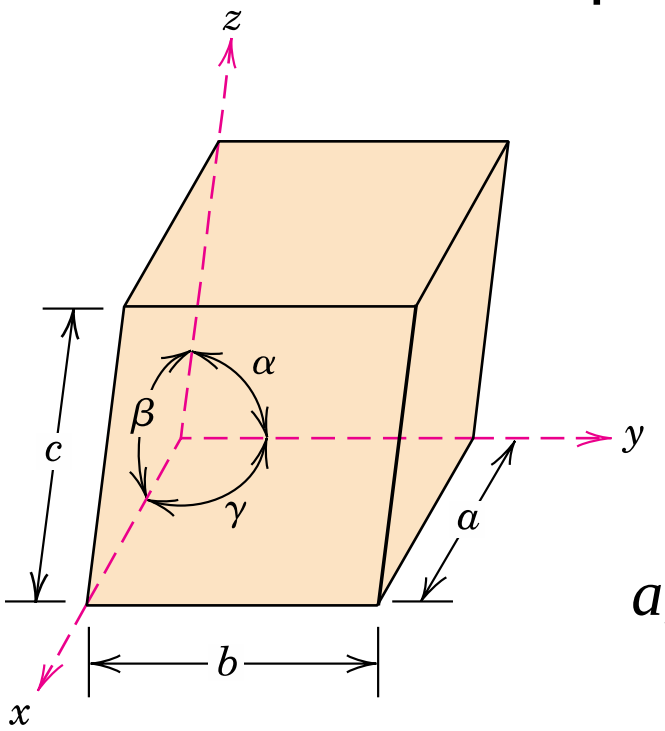
Adapted from Fig. 3.11(b), *Callister & Rethwisch 9e.*

Materials and Packing (cont'd)

- When describing crystalline structures, atoms (or ions) are thought of as solid spheres having well defined diameters.
- This is termed the *atomic hard-sphere model* in which spheres representing nearest-neighbour atoms touch one another
- Sometimes, the term lattice is used in the context of crystal structures
- Lattice means a three-dimensional array of points coinciding with atom positions (or sphere centers)

Crystal Systems

Unit cell: smallest repetitive volume which contains the complete lattice pattern of a crystal.



7 crystal systems

14 crystal lattices

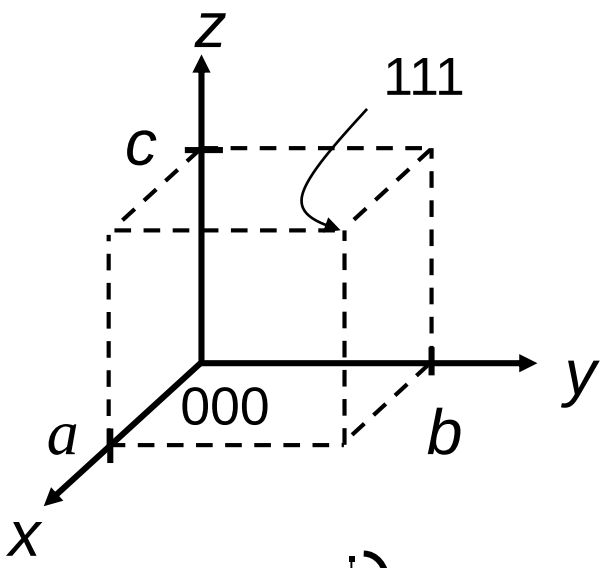
a, *b*, and *c* are the lattice constants

- The unit cell is the basic structural unit or building block of the crystal structure and defines the crystal structure by virtue of its geometry and atom positions within

Point Coordinates

- Labelling conventions have been established in which three numbers or indices are used to designate point locations, directions, and planes
- The basis for determining index values is the unit cell, with a right-handed coordinate system consisting of three (*x, y,* and *z*) axes
- The position of any point located within a unit cell may be specified in terms of its coordinates as fractional multiples of the unit cell edge lengths

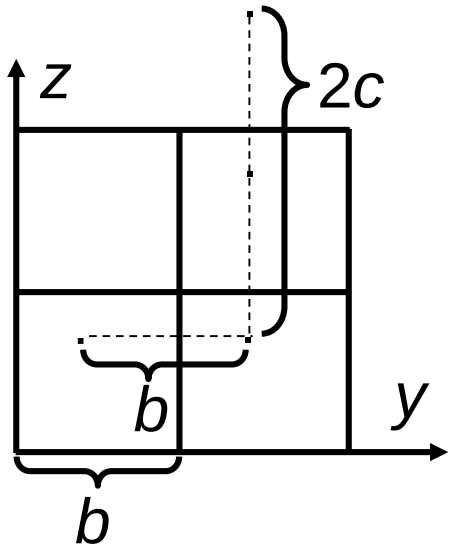
Point Coordinates



Point coordinates for unit cell center are

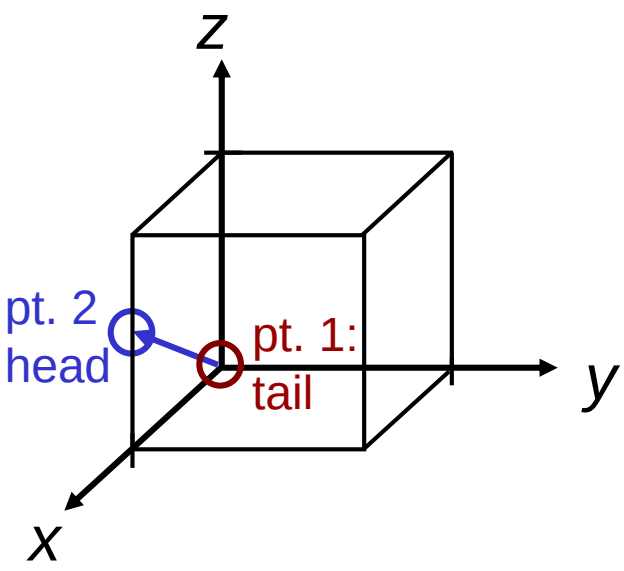
a/2, *b*/2, *c*/2 ½ ½ ½

Point coordinates for unit cell corner are 111



Translation: integer multiple of lattice constants → identical position in another unit cell

Crystallographic Directions



Algorithm

1. Determine coordinates of vector tail, pt. 1: x_1, y_1, & z_1; and vector head, pt. 2: x_2, y_2, & z_2.
2. Tail point coordinates subtracted from head point coordinates.
3. Normalize coordinate differences in terms of lattice parameters a, b, and c:

$$\frac{x_2 - x_1}{a} \quad \frac{y_2 - y_1}{b} \quad \frac{z_2 - z_1}{c}$$

4. Adjust to smallest integer values
5. Enclose in square brackets, no commas

[*uvw*]

ex:

pt. 1 $x_1 = 0, y_1 = 0, z_1 = 0$

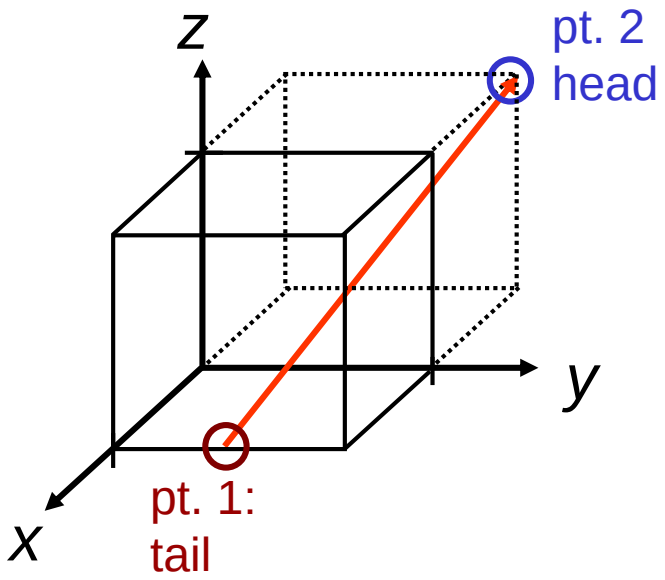
pt. 2 $x_2 = a, y_2 = 0, z_2 = c/2$

$$\frac{a-0}{a} \quad \frac{0-0}{b} \quad \frac{c/2-0}{c}$$

=> 1, 0, 1/2 => 2, 0, 1

=> [201]

Crystallographic Directions



Example 2:

pt. 1 $x_1 = a$, $y_1 = b/2$, $z_1 = 0$

pt. 2 $x_2 = -a$, $y_2 = b$, $z_2 = c$

$$\frac{-a-a}{a} \quad \frac{b-b/2}{b} \quad \frac{c-0}{c}$$

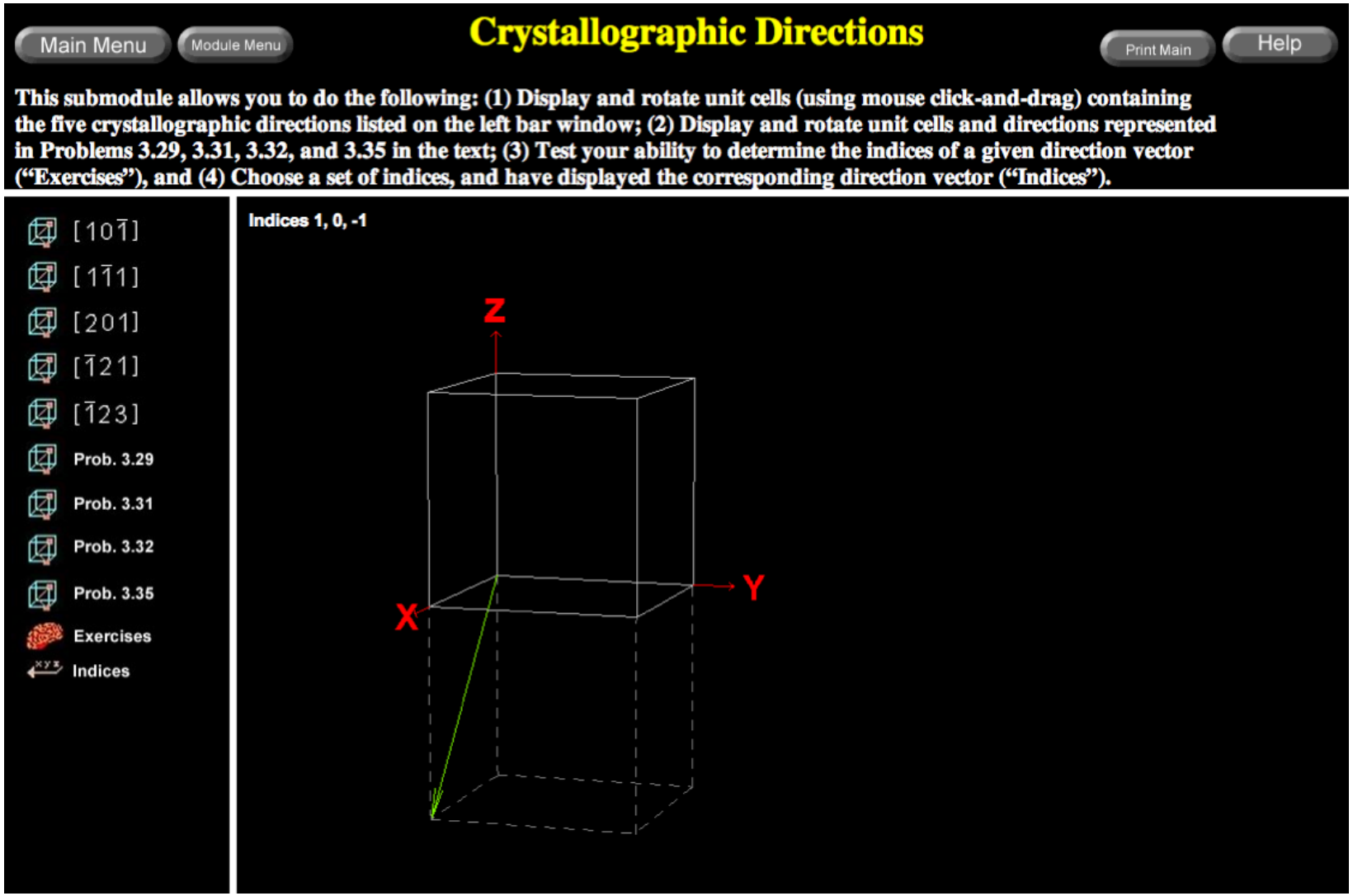
=> -2, 1/2, 1

Multiplying by 2 to eliminate the fraction

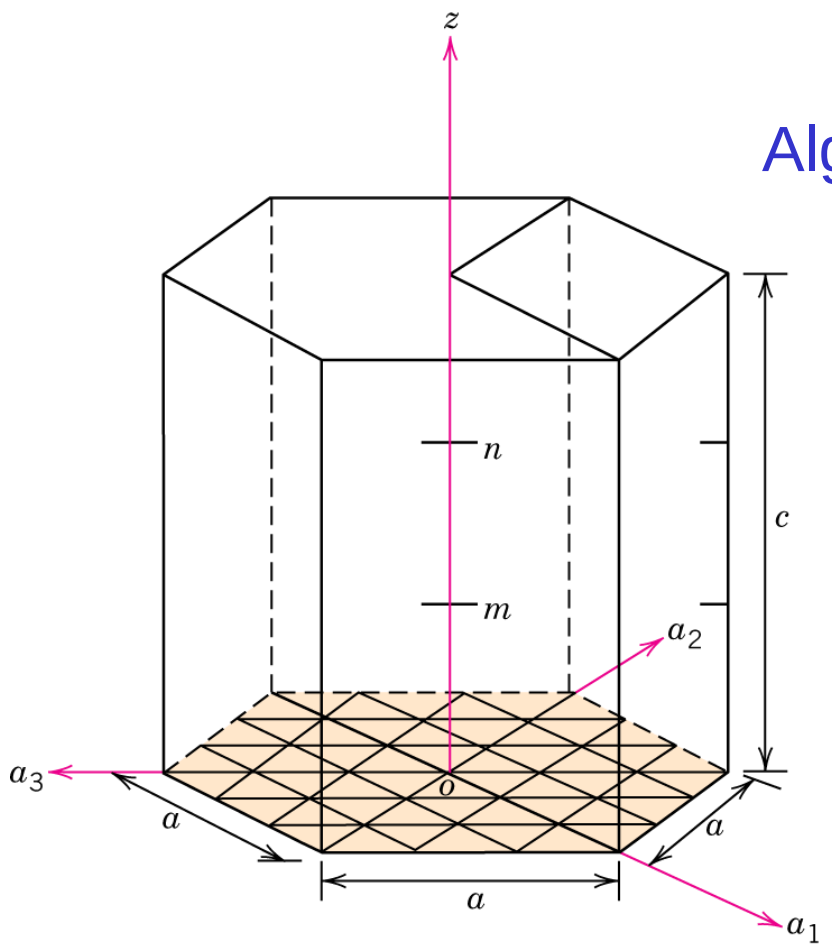
-4, 1, 2 => $[\bar{4}12]$ where the overbar represents a negative index

families of directions <*uvw*>

VMSE Screenshot – [10$\bar{1}$] Direction



Drawing HCP Crystallographic Directions (i)

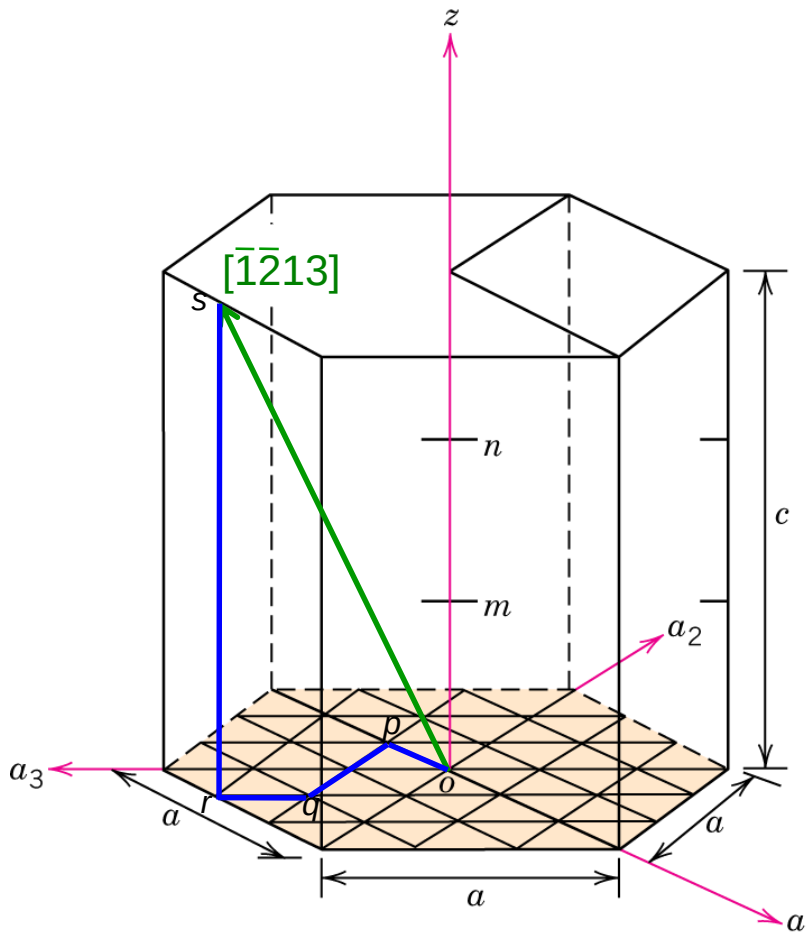


Algorithm (Miller-Bravais coordinates)

1. Remove brackets
2. Divide by largest integer so all values are ≤ 1
3. Multiply terms by appropriate unit cell dimension a (for a_1, a_2, and a_3 axes) or c (for z-axis) to produce projections
4. Construct vector by placing tail at origin and stepping off these projections to locate the head

Drawing HCP Crystallographic Directions (ii)

- Draw the $[\bar{1}\,\bar{2}13]$ direction in a hexagonal unit cell.

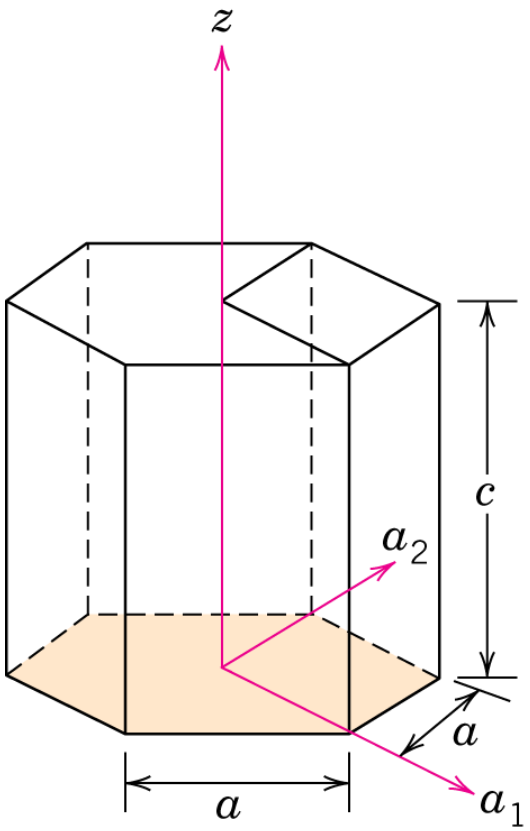


Algorithm	a_1	a_2	a_3	z
1. Remove brackets	-1	-2	1	3
2. Divide by 3	$-\frac{1}{3}$	$-\frac{2}{3}$	$\frac{1}{3}$	1
3. Projections	$-\frac{a}{3}$	$-\frac{2a}{3}$	$\frac{a}{3}$	c

4. Construct Vector
   - start at point *o*
   - proceed $-a/3$ units along a_1 axis to point *p*
   - $-2a/3$ units parallel to a_2 axis to point *q*
   - $a/3$ units parallel to a_3 axis to point *r*
   - *c* units parallel to *z* axis to point *s*

$[\bar{1}\bar{2}13]$ direction represented by vector from point *o* to point *s*

Determination of HCP Crystallographic Directions (ii)



Algorithm

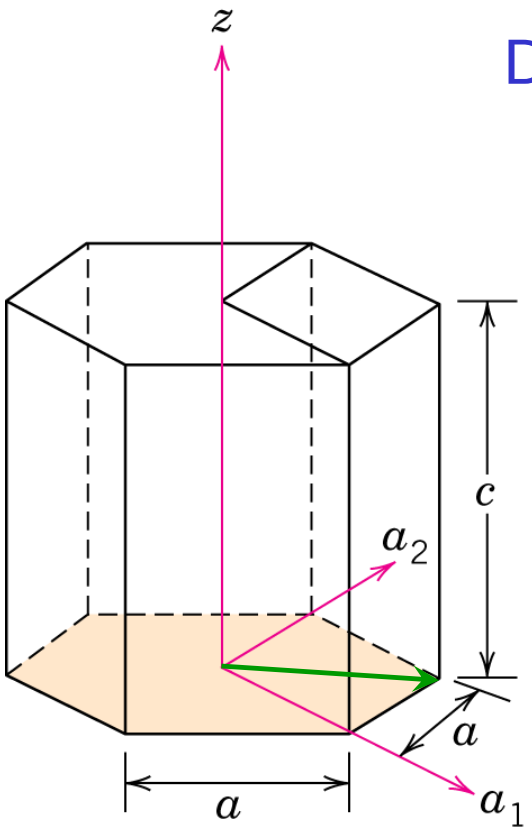
1. Determine coordinates of vector tail, pt. 1: x_1, y_1, & z_1; and vector head, pt. 2: x_2, y_2, & z_2. in terms of three axis (a_1, a_2, and z)
2. Tail point coordinates subtracted from head point coordinates and normalized by unit cell dimensions a and c
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas, for three-axis coordinates [$u'v'w'$]
5. Convert to four-axis Miller-Bravais lattice coordinates using equations below:

$$u = \frac{1}{3}(2u - v) \qquad v = \frac{1}{3}(2v - u)$$

$$t = -(u + v) \qquad w = w$$

6. Adjust to smallest integer values and enclose in brackets [*uvtw*]

Determination of HCP Crystallographic Directions (ii)



Determine indices for green vector

Example		a_1	a_2	z
1.	Tail location	0	0	0
	Head location	a	a	$0c$
2.	Normalized	1	1	0
3.	Reduction	1	1	0
4.	Brackets		[110]	

5. Convert to 4-axis parameters

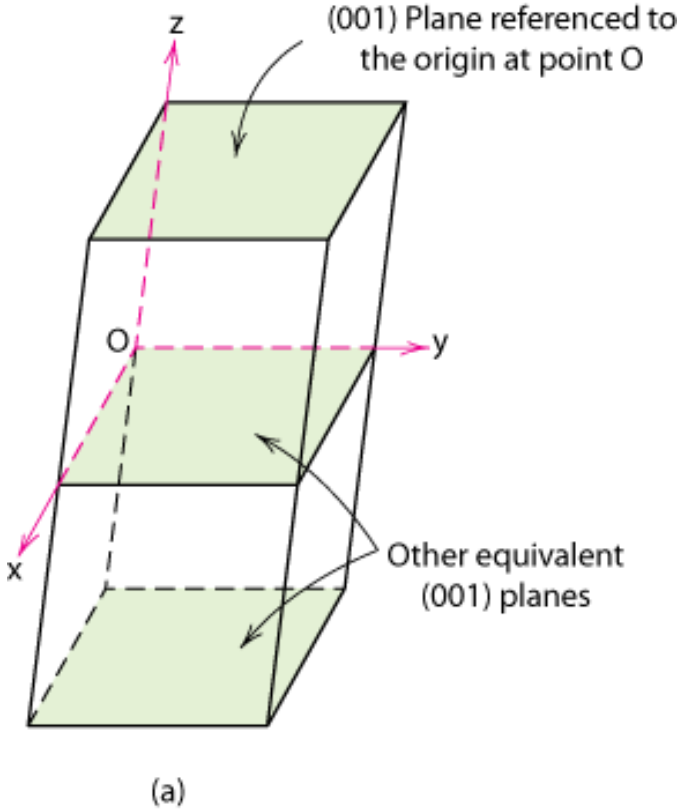
$$u = \frac{1}{3}[(2)(1) - (1)] = \frac{1}{3} \qquad v = \frac{1}{3}[(2)(1) - (1)] = \frac{1}{3}$$

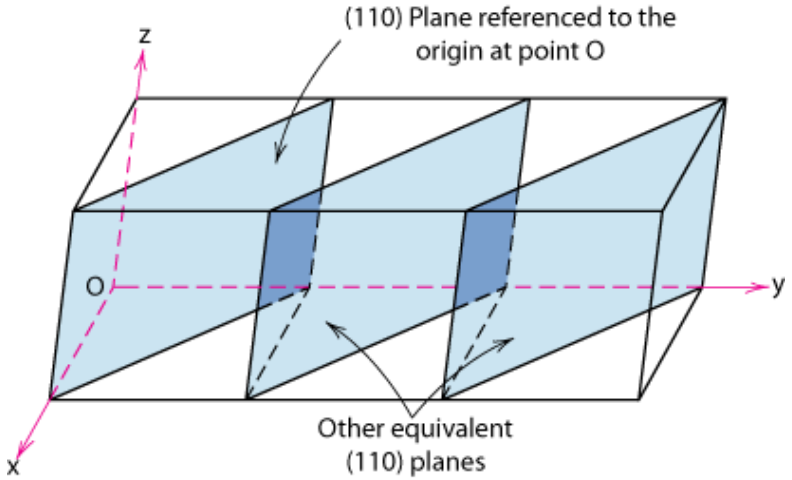
$$t = -(\frac{1}{3} + \frac{1}{3}) = -\frac{2}{3} \qquad w = 0$$

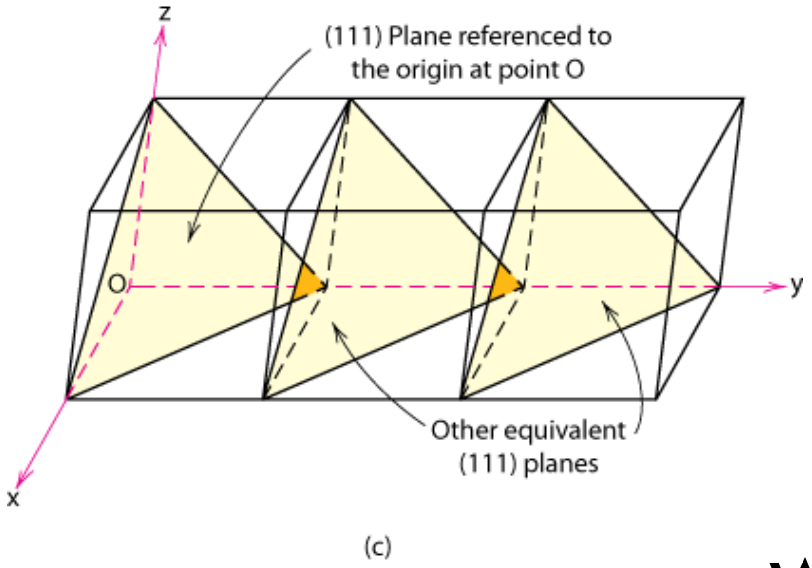
6. Reduction & Brackets

1/3, 1/3, -2/3, 0 => 1, 1, -2, 0 => $[11\bar{2}0]$

Crystallographic Planes







Adapted from Fig. 3.7, *Callister & Rethwisch 9e.*

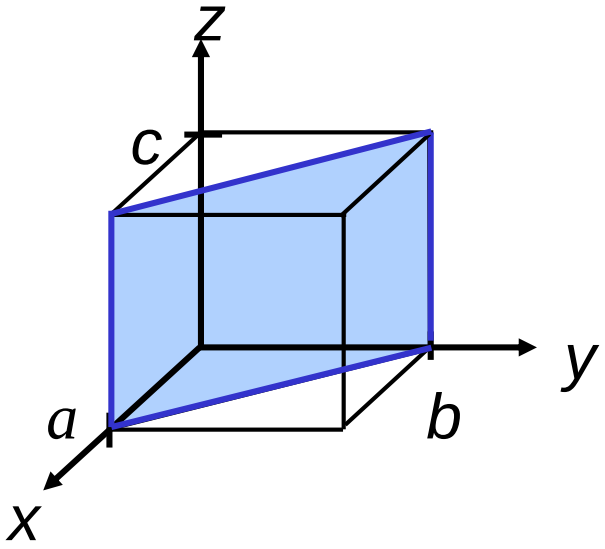
Crystallographic Planes

- Miller Indices: Reciprocals of the (three) axial intercepts for a plane, cleared of fractions & common multiples. All parallel planes have same Miller indices.

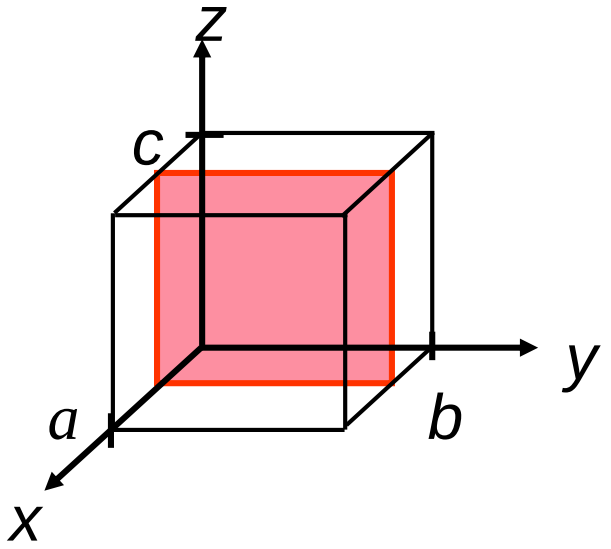
- Algorithm
  1. Read off intercepts of plane with axes in terms of a, b, c
  2. Take reciprocals of intercepts
  3. Reduce to smallest integer values
  4. Enclose in parentheses, no commas i.e., (*hkl*)

Crystallographic Planes

example		*a*	*b*	*c*
1.	Intercepts	1	1	∞
2.	Reciprocals	1/1	1/1	1/∞
		1	1	0
3.	Reduction	1	1	0
4.	Miller Indices	(110)		

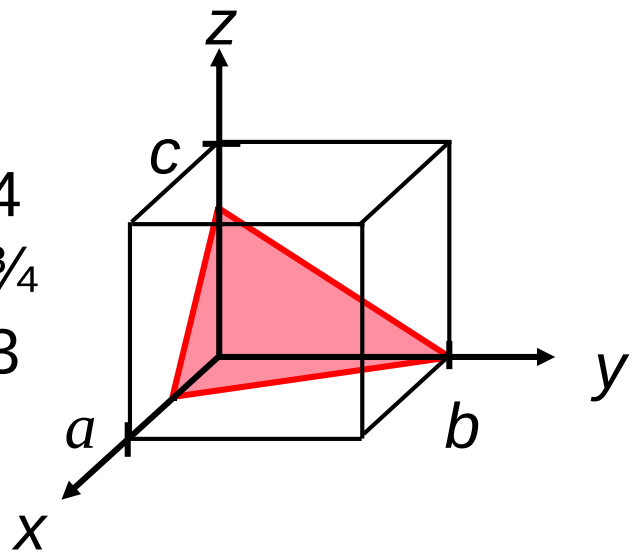


example		*a*	*b*	*c*
1.	Intercepts	1/2	∞	∞
2.	Reciprocals	1/½	1/∞	1/∞
		2	0	0
3.	Reduction	2	0	0
4.	Miller Indices	(100)		



Crystallographic Planes

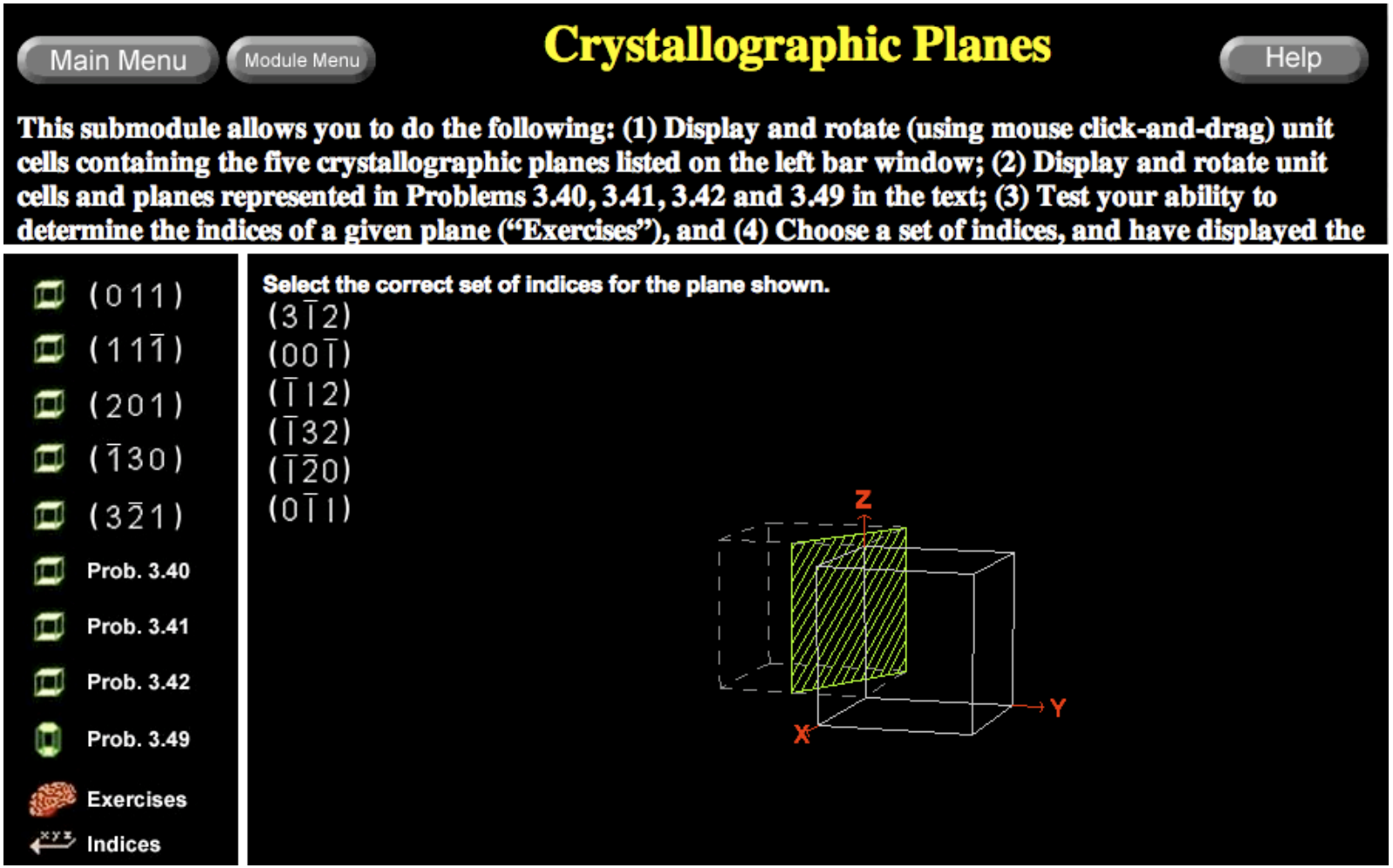
example	*a*	*b*	*c*
1. Intercepts	1/2	1	3/4
2. Reciprocals	1/½	1/1	1/¾
	2	1	4/3
3. Reduction	6	3	4
4. Miller Indices	(634)		



Family of Planes {*hkl*}

Ex: {100} = (100), (010), (001), ($\bar{1}$00), (0$\bar{1}$0), (00$\bar{1}$)

VMSE Screenshot – Crystallographic Planes



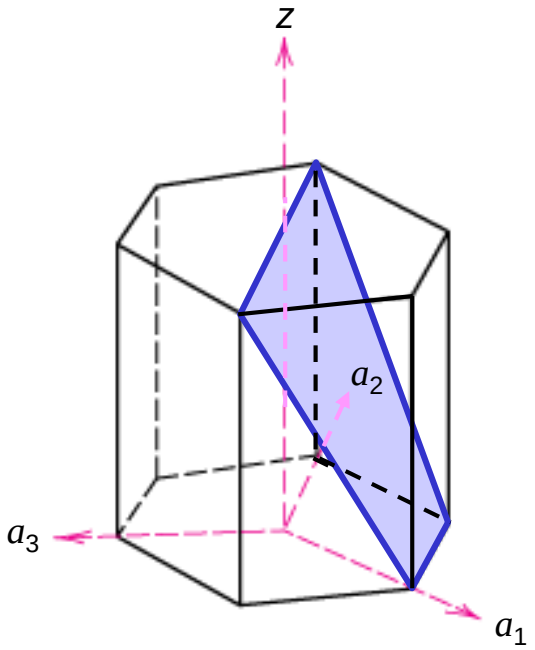
Additional practice on indexing crystallographic planes

Crystallographic Planes (HCP)

- In hexagonal unit cells the same idea is used

example	a_1	a_2	a_3	c
1. Intercepts	1	∞	-1	1
2. Reciprocals	1	$1/\infty$	-1	1
	1	0	-1	1
3. Reduction	1	0	-1	1

4. Miller-Bravais Indices $(10\bar{1}1)$



Adapted from Fig. 3.8, *Callister & Rethwisch 9e.*

Crystallographic Planes

Draw (100) and (111) crystallographic planes for Fe.

Summary

- Atoms may assemble into crystalline or amorphous structures.
- Crystallographic points, directions and planes are specified in terms of indexing schemes. Crystallographic directions and planes are related to atomic linear densities and planar densities.

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: